32-cell monitor interface

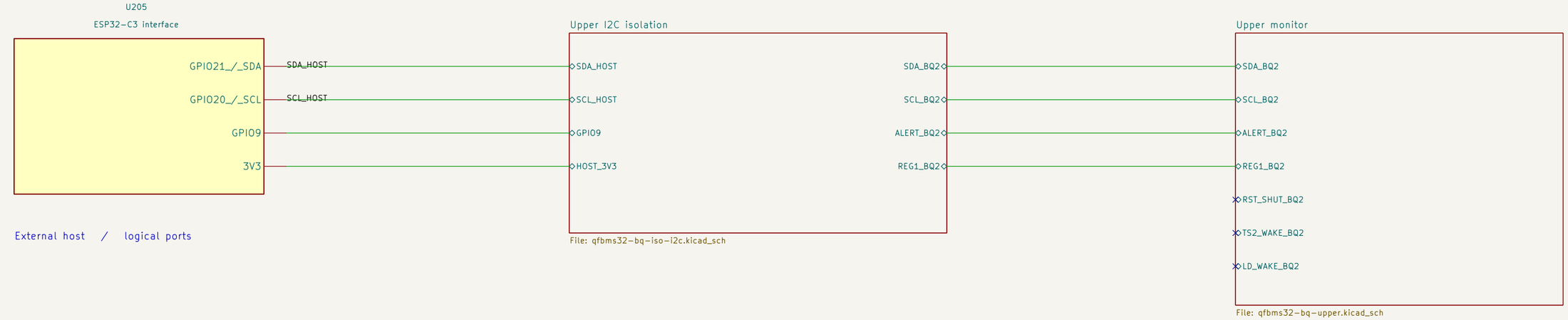
Charge-only BMS / 100 kHz I2C / ESP32-C3 host

HOST DOMAIN / CELL0

UPPER DOMAIN / CELL16

U204 TMUX154E > U_ISO ISO1640

Domain boundary inside ISO1640

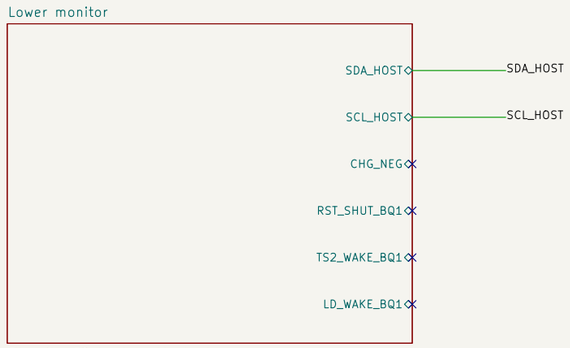


External host / logical ports

File: qfbms32-bq-iso-i2c.kicad_sch

File: qfbms32-bq-upper.kicad_sch

Cells 17-32 | upper address 0x08



File: qfbms32-bq-lower.kicad_sch

Cells 1-16 | lower address 0x08 at reset

Startup sequence

1 GPIO9 high opens U204. Configure lower BQ on the direct host bus.
2 Verify lower BQ at 0x10.
3 Drive GPIO9 low to admit upper BQ at 0x08 through ISO1640.

OPEN Lower BQ remains connected with its rail off. See lower sheet.
OPEN Both TS2 wake drivers require local VSS switching with REG1 off.

Schematic increment only. qfbms32.kicad_pcb is not updated from this file.

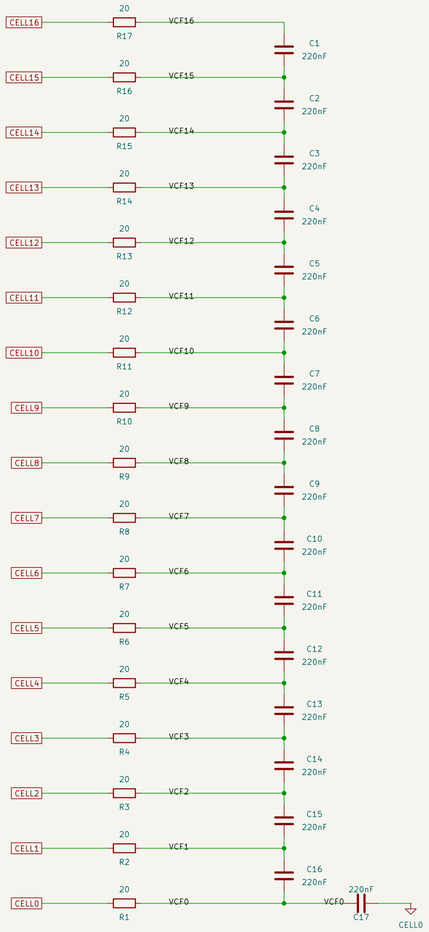
Sheet: /
File: qfbms32-bq-monitors.kicad_sch

Title: QFBMS32 BQ monitor sheets

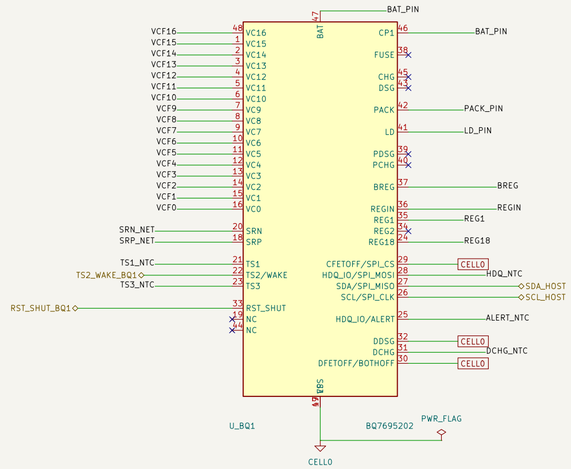
Size: A3	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/4

Lower monitor / cells 1 to 16 / reference CELL0

01 Cell input filters

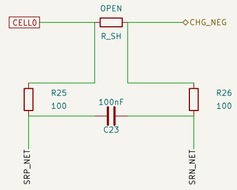


02 Monitor pinout



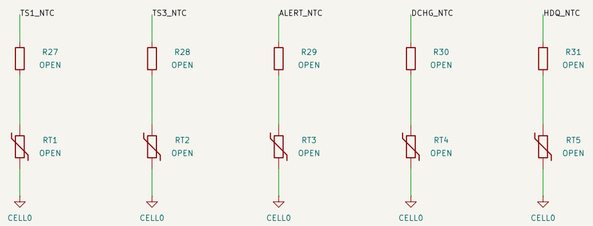
OPEN SDA/SCL connect directly to host. Lower powered-off isolation requires a hardware decision; a second isolator is a candidate.

03 Current sense



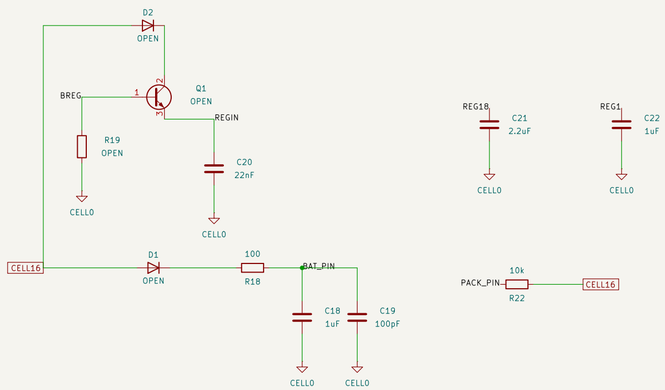
OPEN Confirm Kelvin pickup, shunt value and firmware sign/gain.

04 Temperature inputs



OPEN Firmware assumes 10 kOhm / beta 3380 and 500 ohm series correction.

05 BAT and local regulators



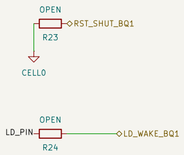
OPEN Select stack-rated diodes, REGO transistor and passive ratings. Check REGIN headroom and dissipation.

06 I2C pull-ups



OPEN Select pulls at 100 kHz. Check rail limits, sink current and rise time for this segment.

07 Wake interface



OPEN TS2 wake driver: pulse to CELL0, then release.
Candidate: optical output powered from the host LED side.
Verify leakage, pulse width and GPIO availability.

Charge-only monitor sheet. Do not import this netlist into qfbms32.kicad_pcb.

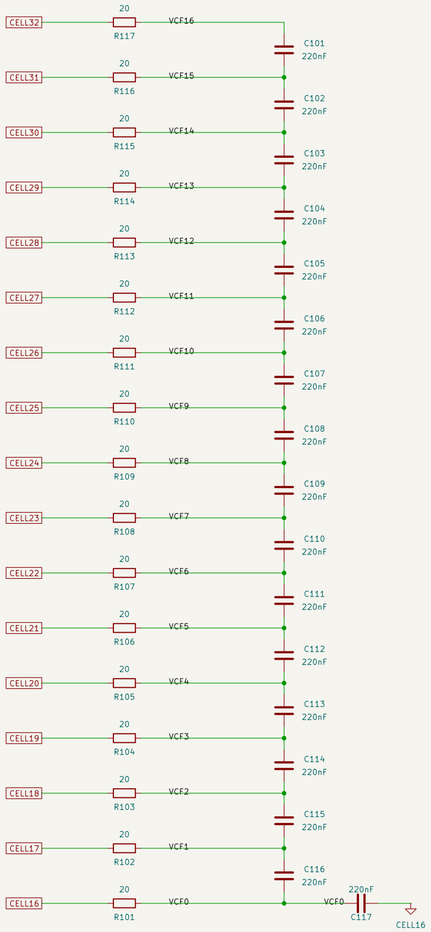
Sheet: /Lower monitor/
File: qfbms32-bq-lower.kicad_sch

Title: QFBMS32 lower monitor U_BQ1

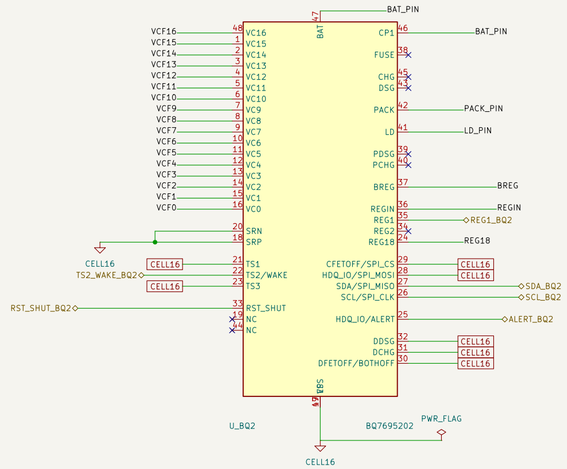
Size: A2	Date: 2026-11-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 2/4

Upper monitor / cells 17 to 32 / reference CELL16

01 Cell input filters



02 Monitor pinout



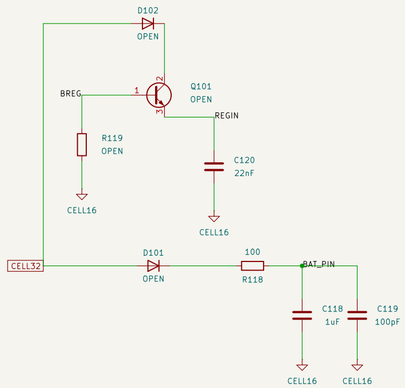
03 Current sense

SRP (18) and SRN (20) share upper VSS (17).
The current shunt is on the lower monitor.

04 Temperature inputs

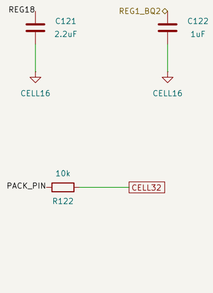
Internal temperature only. Unused sensor inputs return to CELL16.
ALERT terminates on the interface sheet.

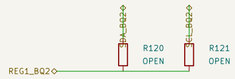
05 BAT and local regulators



OPEN Select stack-rated diodes, REGO transistor and passive ratings. Check REGIN headroom and dissipation.

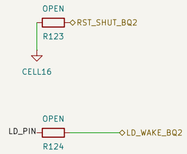
06 I2C pull-ups





OPEN Select pulls at 100 kHz. Check rail limits, sink current and rise time for this segment.

07 Wake interface



OPEN TS2 wake driver: pulse to CELL16, then release. Candidate: optical output powered from the host LED side. Verify leakage, pulse width and GPIO availability.

Charge-only monitor sheet. Do not import this netlist into qfbms32.kicad_pcb.

Sheet: /Upper monitor/
File: qfbms32-bq-upper.kicad_sch

Title: QFBMS32 upper monitor U_BQ2

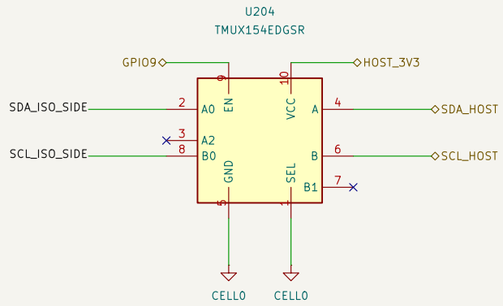
Size: A2	Date: 2026-11-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 3/4

Upper monitor communication

CELL0 host domain

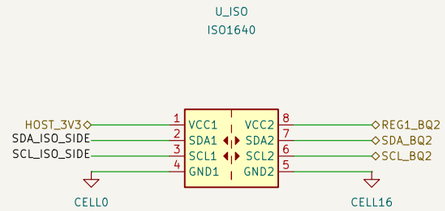
CELL16 upper domain

01 GPIO9-controlled disconnect



GPIO9 high opens both lines. Low selects A0/B0 after the lower address check.

02 ISO1640 domain boundary



OPEN Select ISO1640 orderable 8-D suffix. Check rail margins and sequencing.

03 Host pull-ups and reset bias



OPEN R201/R202: 4.7 kOhm candidate at 100 kHz. Verify rise time, sink current and 80 pF side-1 limit.
OPEN R203: size for reset-high GPIO9 and ESP32-C3 boot strapping.

Upper ALERT stops here; host firmware does not read it.

04 Local bypass



GPIO9 is pulled high at reset. GPIO9 low selects ESP32-C3 download mode.

Sheet: /Upper I2C isolation/
File: qfbms32-bq-iso-i2c.kicad_sch

Title: QFBMS32 upper I2C isolation

Size: A3	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/4